

Is a passing bay worth a transporter?

Fleet-network co-design: the road-vehicle exchange-rate theorem, a retrofit menu that includes control granularity, and vehicle type under road occupancy

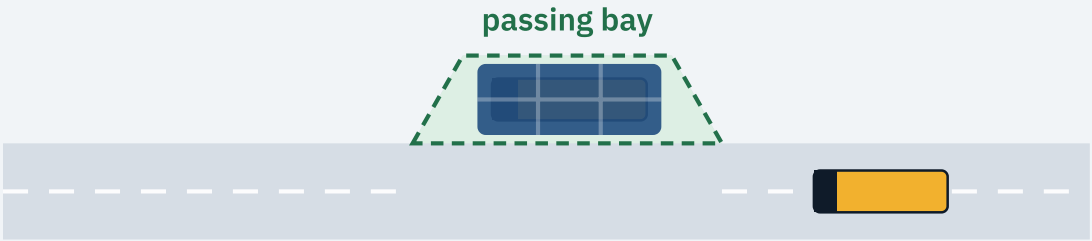
Is a passing bay worth a transporter? Fleet-network co-design for road-occupying transport (working title)

Chunrui Na

Harbin Engineering University · Nanyang Technological University

Established: the capacity intervals, the system bound and two columns of dual prices; a less conservative deadlock-free rule is being compared, and the first exchange-rate table follows

exchange rate = dual price ÷ daily capacity per vehicle



On a single lane, the loaded carrier pulls into the bay and an empty one passes

≈ ?



how many vehicles is it worth?



The sub-study in one minute

1

The question

To expand capacity where the network binds: buy vehicles or fix roads, large vehicles or small? We price each road change in vehicles.

2

The dividing rule

Plateau / bound near 1: the bound is tight, its dual prices are the exchange rate, and fixing the road pays. Far below 1: the rule is the bottleneck, so change the orchestration first.

3

Results so far

Over 10 seeds: segmenting Okpo's dock road lifts the upper edge (the bound) by +8.0% but the plateau by only +0.3%. In the erection mix the dock binds (0.87): +76.5% and +8.3%, and the ratio falls to 0.53.

4

Next

Two columns of dual prices and the route slack are in hand; a less conservative deadlock-free rule is being compared; then retrofit simulations build the first exchange-rate table, and floating docks follow.

Where the dock mouth dominates, a road that can be used in sections may be worth more than a wider one.

Chapter 5's question: when we spend to expand capacity, vehicles or roads, large vehicles or small?

When loaded vehicles hold the road exclusively, what determines the capacity of a shipyard block-transport system? How should fleet, orchestration and network be designed together?



What the chapter takes in

- **Chapter 3:** the network bound and its noise-free system version, the route-free bound, and two columns of dual prices
- **Chapter 4:** the best deadlock-free rule, the interval's lower edge and the inner evaluator; now whole-route reservation, being re-checked
- **The fleet-sizing paper:** vehicle types, the team-transport model and vehicle prices from 33 tender awards

What the chapter delivers

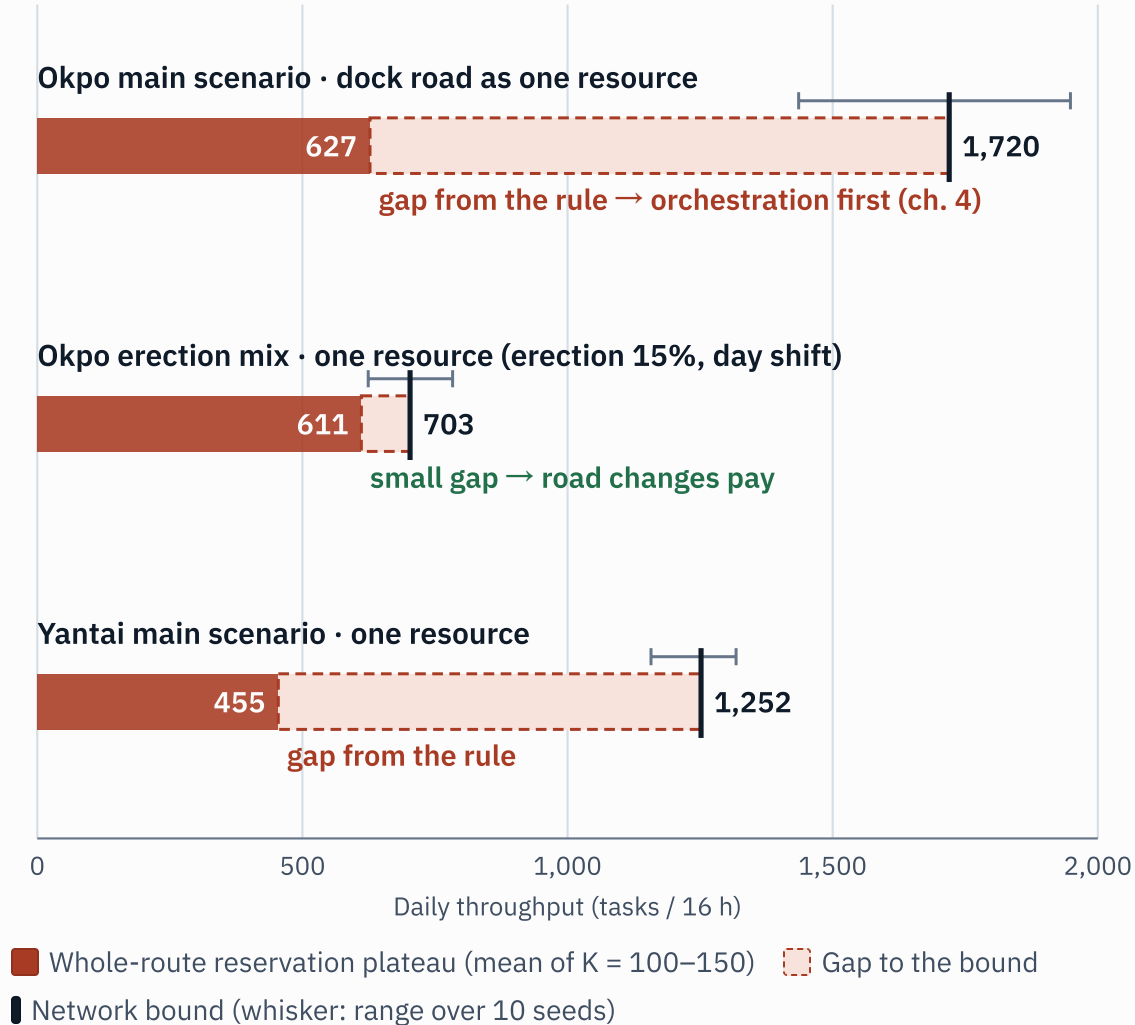
- The rule dividing orchestration from road changes, and the first road–vehicle exchange-rate table
- A proof of the exchange-rate theorem and critical-element identification; a retrofit menu with control granularity
- A bilevel co-design algorithm; the vehicle-type result; upgrade values handed to chapter 6 for cross-yard prediction

The claim, hypotheses and theorem of this chapter: exchange rate, upgrade value, vehicle type

#	STATEMENT (VERSION OF 9 OCT 2026)	HOW THE CHAPTER DELIVERS IT
1	Road–vehicle exchange rate: where the bound is tight, upgrading critical elements can replace vehicles, at a rate equal to the dual price converted to vehicles. When the plateau is far below the bound, change the orchestration first. This is its boundary with the traffic-ordering claim	The dividing rule is the chapter's backbone; 10-seed simulations give two examples (main scenario, erection mix)
2	Upgrade-value hypothesis: where the network binds, one upgrade on a critical element is worth at least one vehicle; critical elements can be identified by dual prices. Upgrades to choose from: segmented blocking, dock-front holding bays, passing bays, one-way loops, widening, a second entrance. Criterion: the difference in vehicles needed before and after the upgrade in simulation, compared with the dual-price prediction. The two are expected to agree only when plateau / bound is close to 1. Holds: the exchange rate holds. Fails: the capacity bottleneck is orchestration, and chapter 4 is strengthened.	Retrofit simulations (after the rule comparison): 10 seeds, full K grid; difference in vehicles needed at whole-yard volume; dual predictions from the two computed columns
3	Vehicle-type hypothesis: large vehicles occupy the road both loaded and empty; small ones can yield when empty, but teaming up for a lift costs extra empty runs. Case: multi-vehicle transport of giant blocks for floating docks. Holds: the optimal fleet mix changes with how strongly the network binds. Fails: vehicle type and network can be decided separately.	Floating-dock scenario on the Okpo map with floating docks (from the rule comparison on); the fleet-sizing paper's vehicle types and prices
4	Exchange-rate theorem: given by the dual prices; only upgrades on binding constraints have value. For new links, use the route-free bound Status: proof in writing	Proved in this study; the route-free bound (draft proof being rewritten) and two columns of dual prices are computed
From dispatching to traffic ordering (the other side): where the network binds, how much of the bound is reachable is decided by traffic orchestration: release order, waiting points, reservation mode, block granularity. Naive dispatching deadlocks.		

Source: this study's claims and hypotheses (version of 9 Oct 2026), quoted verbatim on the left; the right column describes how we carry them out.

Orchestration or roads: how far the plateau sits below the bound decides



36%

36% Far below 1: the rule is the bottleneck

Raising the bound leaves throughput unchanged: improve orchestration first (chapter 4 raises the lower edge). Example: segmenting Okpo's dock-1 road lifts the bound by +8.0%, the reservation plateau by only +0.3%; Yantai's main scenario also sits at 36%.

87%

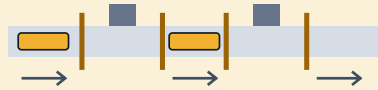
87% Near 1: the bound is tight

Dual prices are the exchange rate; fix the binding element first. Example: in the erection mix the dock-1 road binds (86%, completed-mix caliber); segmenting it raises the reservation plateau by +8.3%.

36%

Capacity is an interval [best deadlock-free rule, network bound]; the lower edge is whole-route reservation (best so far, being re-checked). Chapter 4 raises it, chapter 3's tighter bounds lower the upper edge, and their ratio decides where chapter 5's money goes.

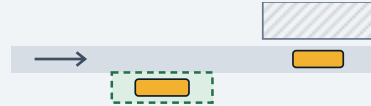
Retrofit menu: six changes, one purely in control granularity



CONTROL GRANULARITY

Segmented blocking (with direction locks)

A long road is cut at its stops into several resources, with same-direction following. Routes stay as they are, so the network bound still applies.



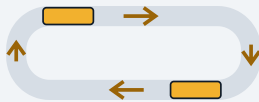
Dock-front holding bay

Vehicles waiting for the crane stand off the road and stop blocking the way to neighbouring stops. Where vehicles wait is not in the LP, so the value is read from simulation.



Passing bay

A wider stretch on a single lane where two vehicles pass. It changes routes, so the route-free bound prices it.



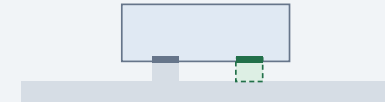
One-way loop

Turns head-on conflicts into same-direction following. Directions and shortest paths change, so the route-free bound is needed.



Widening

Width capacity rises in steps: +1 m on the 9 m dock-1 road leaves the bound unchanged; at 12 m the two narrowest vehicles pass.



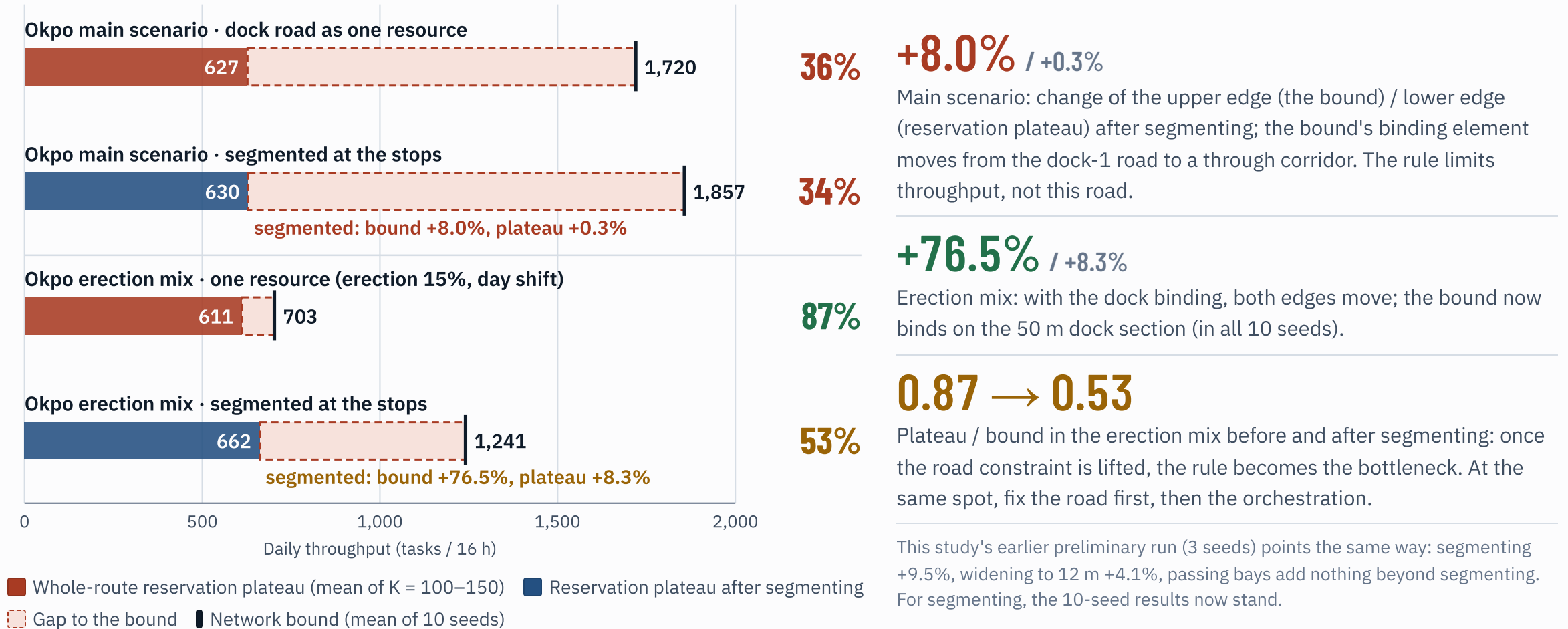
Second entrance

Adds a door where a door binds, sharing out the loading time held at the node. It adds a link, so the route-free bound is needed.

Control granularity is hidden capacity: segmented blocking needs no construction, only a new division of resources and release of vehicles. Block granularity belongs to traffic ordering and segmented blocking to the upgrade menu, so it sits on the boundary.

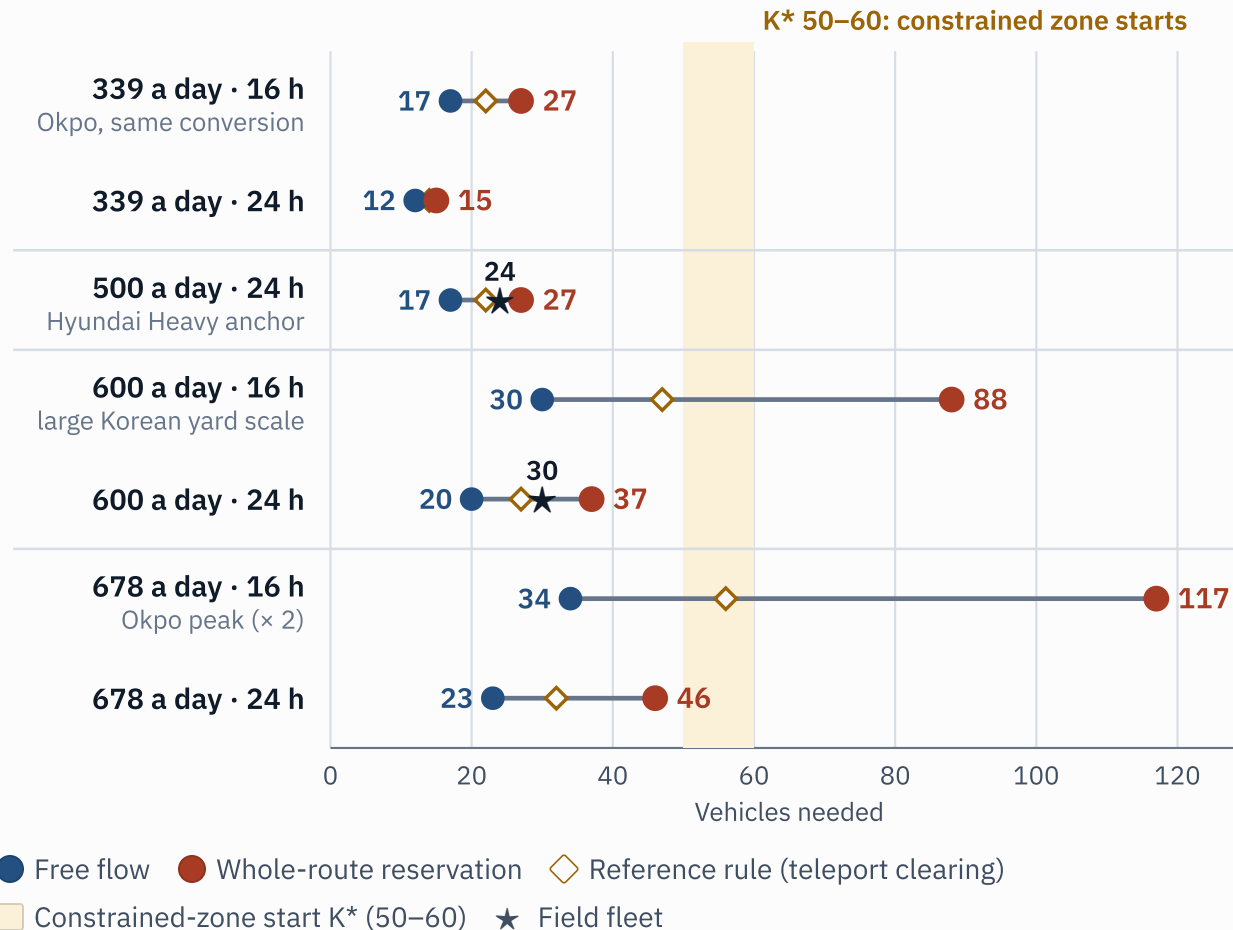
Migration sources (full texts): a railway block section is held by one train (Abril et al. 2008); shorter blocks use less capacity, best length open (Goverde et al. 2013); UIC 406 (Landex et al. 2008); a mine's single lane with passing bays, run as block sections (Eustace et al. 2021).

Two examples of segmenting: the main scenario lifts only the upper edge; with the dock binding, both edges rise



Sources: simulation in this study. Conditions: saturated demand, 4 days, 10 seeds, whole-route reservation; block boundaries 25 m either side of each stop on the dock road (a scenario setting); segmented is Okpo's main scenario (imagery check, 9 Oct 2026), the whole road the comparison; Okpo map before the floating docks and branch roads. The earlier 3-seed figures are this study's preliminary run.

Vehicles needed at whole-yard volume: the real scale of “worth so many vehicles”



17 → 27

Okpo's own 339 a day, 16 h: free flow → whole-route reservation; at 24 h, 12 → 15. Both below the constrained-zone start K* (50–60).

34 → 117

Okpo's peak, 678 a day (× 2), 16 h: inside the network-constrained zone; under 24 h, 23 → 46, at the zone's lower edge.

24 · 30

Field fleets: Hyundai Heavy, ~500 moves a day, 24 vehicles, 24 h; a large Korean yard, ~600 a day, ~30 vehicles (shift not stated, read at 24 h). Both lie between free flow and reservation at 24 h.

16 h or 24 h, daily mean or peak: this main caliber of the network-bound hypothesis is fixed after the rule comparison, so both are shown. The exchange rate addresses the network-constrained zone.

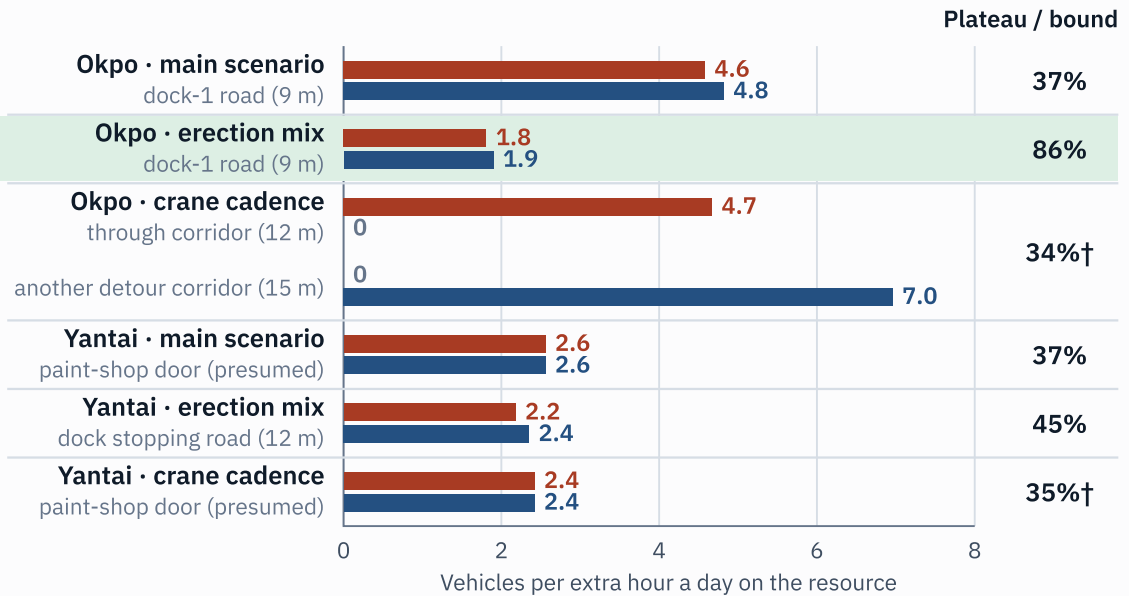
Sources: simulation in this study (integer K, 10 seeds; vehicles needed = smallest K with mean throughput $\geq 0.99 \times$ daily volume; dock road as one resource, segmented within one vehicle; K* from the same runs). 339 = Okpo's steel use, same conversion (yearly 294–363; Hanwha system's direct anchor 350); peak × 2. Field anchors: Park and Seo 2006; Shen et al. 2018. Okpo map before floating docks.

The exchange-rate theorem: the bound's dual prices in vehicles; only binding elements are worth changing

$\pi_r = \partial(\text{bound}) / \partial H_r$ extra tasks a day per extra hour a day on resource r

$e_r = \pi_r \div (H / \bar{c})$ in vehicles; $H / \bar{c}$ = one vehicle's daily capacity

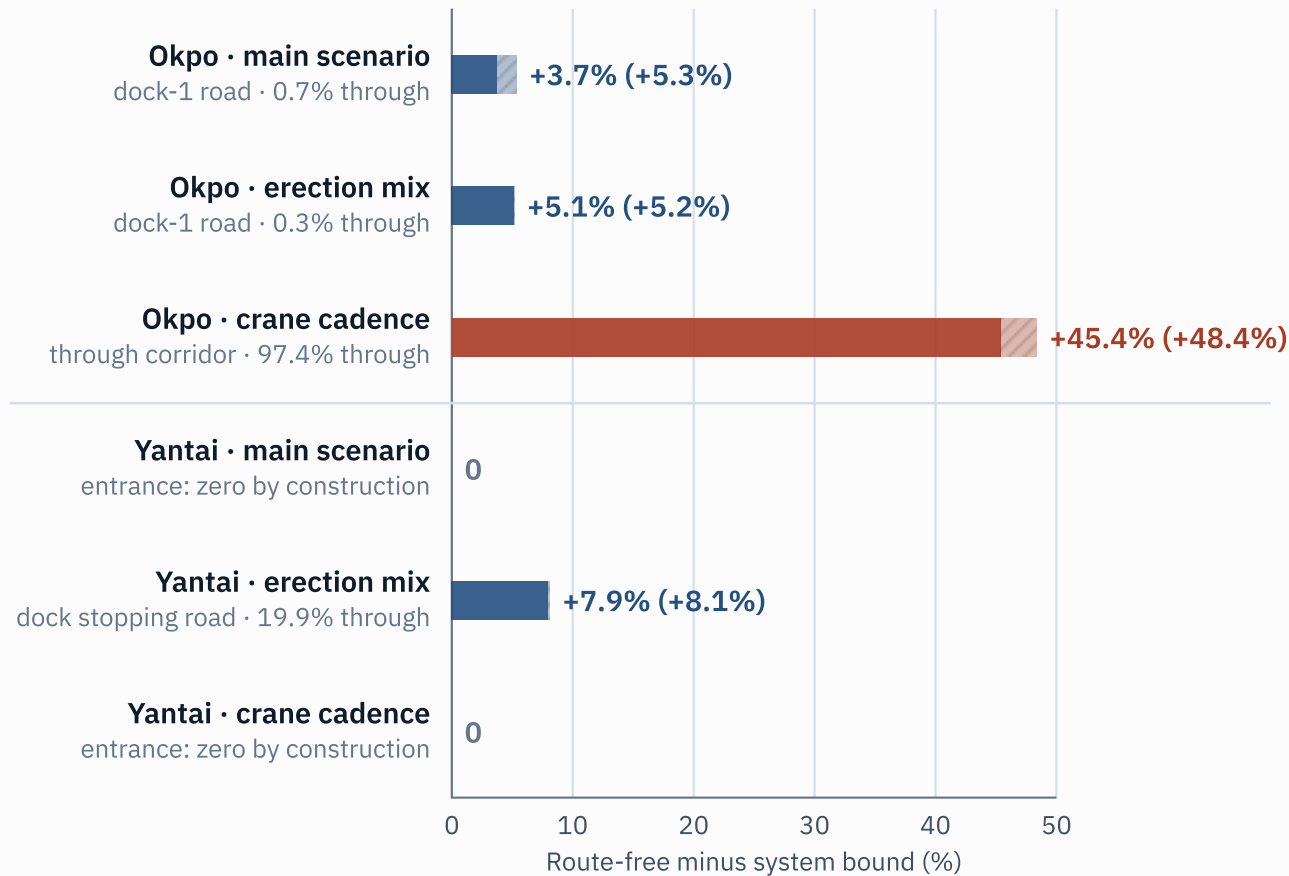
- **Only upgrades on binding constraints have value:** a slack resource has a zero dual price.
- **With plateau / bound near 1, e_r is the exchange rate;** far below 1, it bounds an upgrade's value from above.
- **Prices are marginal and the solution degenerate:** the two doors of Yantai's paint shop (presumed) are full at once; changing one adds only 2.3 a day, both together +144 to first order; the retrofit simulations resolve combined changes.
- **Two columns side by side:** read the system-bound column under shortest-path dispatch and the route-free column when re-routing is allowed (chapter 4's detour orchestration); new links always use the route-free bound.



■ Shortest-path dispatch (system bound) ■ Re-routing allowed (route-free bound)
† crane cadence on the system caliber; others on the completed mix
Extra tasks a day per extra hour a day on the resource, divided by one vehicle's daily capacity (about 24 at Okpo, 32 at Yantai). The erection mix reads as the exchange rate; elsewhere the plateau is 36–45% of the bound, so the vehicles bound an upgrade's value from above.

Sources: the theorem as stated in this study (the exchange-rate theorem, proof in writing); this study's system-bound and dual-price computations (system-bound and route-free columns; resources with a positive price among the top 10 of either; plateau / bound on the completed-mix caliber, † crane cadence on the system caliber); $H / \bar{c}$ = fleet bound ÷ K in the same scenario's simulation. Crane cadence = crane count from the map, erection fixed in absolute terms (gantry-crane spans and erection volume). Okpo map before the floating docks and branch roads; Yantai map.

Route slack in the bound: the most re-routing can add, and where road changes and orchestration substitute



■ Lower: back-substituted feasible value – system bound ■ Up to route-free – system bound

A through corridor (Okpo, crane cadence)

97% of its traffic passes through, and the detour keeps 22% spare at 235 m longer on average. In the shortest-path column a 10% shorter occupancy is worth +198 a day (≈ 8.3 vehicles); in the re-routing column, 0: re-routing can stand in for changing it.

Dock stopping roads and entrances

Dock stopping roads (Okpo's dock-1 road, Yantai's dock road) carry little through traffic; their slack comes mainly from entering the stop from the other end: +3.7% to 7.9%. An entrance has zero slack by construction; only the door itself can change (door granularity, next step).

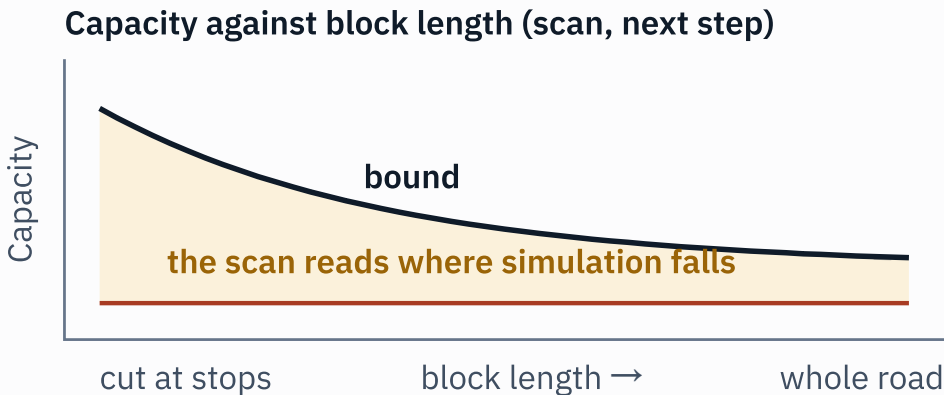
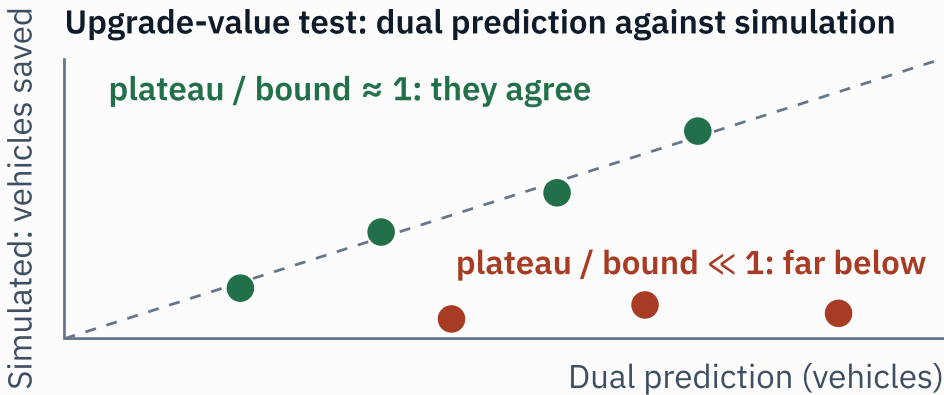
Route slack is a difference of two bounds: the most chapter 4's detour orchestration could recover; what it does recover comes from that orchestration's plateau. The three readings are proposed wording; the retrofit simulations add points.

Sources: this study's system-bound and route-free-bound computations (lower = back-substituted feasible value – system bound, upper = route-free bound – system bound; through share, detour slack and dual prices from the same computation); the draft proof of the route-free bound is being rewritten, and a counterexample search found none. Okpo map before the floating docks and branch roads; Yantai map.

The first exchange-rate table (next step): each change in vehicles, dual prediction against simulation

CHANGE	READINGS SO FAR	DUAL PREDICTION	SIMULATION: Δ VEHICLES NEEDED
Segmented blocking (direction locks)	Erection mix: bound +76.5%, reservation +8.3% (10 seeds)	Dock-1 road: 1.8 vehicles per hour a day	pending
Dock-front holding bay	—	not seen by the LP	pending · holding
Passing bays	Preliminary run: -0.2% beyond segmenting	Route-free bound on the new network	pending
Widening	Preliminary run: 12 m, +4.1%; dock-1 road +1 m: bound unchanged	Next-step pricing; Yantai's dock road +1 m: +127 a day ≈ 4.0 vehicles	pending
Door granularity (Yantai paint shop, presumed)	—	One entrance +2.3 a day; both, first order +144	pending

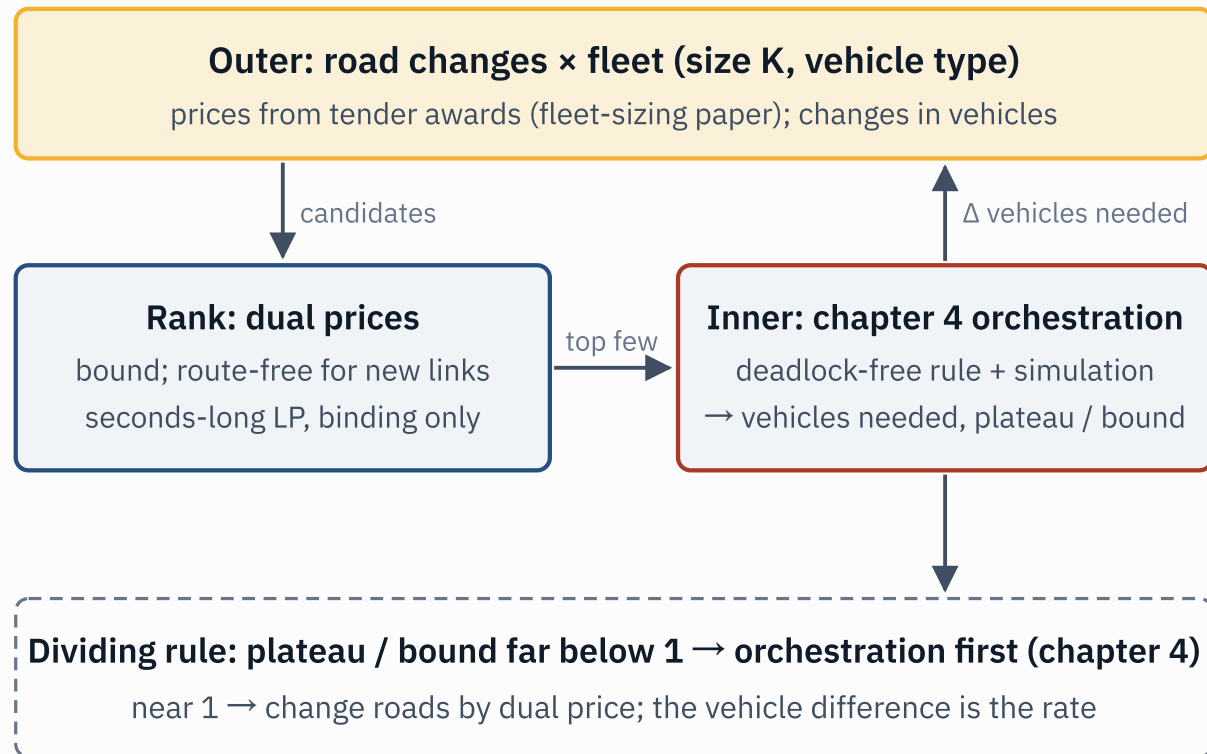
The table's caliber: 10 seeds, full K grid; vehicles needed at Okpo's 339 a day and the 678 peak, 16 h and 24 h side by side; on the less conservative rules and the Okpo map with floating docks and branch roads; conclusions only from rules without teleport clearing.



Both figures are schematics, not data

Sources: this study's research plan (retrofit simulations and the first exchange-rate table; block-length scan; door granularity; dock three-resource network; after the deadlock-free rule comparison, which also switches the Okpo map); readings so far from this study's simulations (10 seeds) and preliminary run (3 seeds); dual predictions from this study's system-bound and dual-price computations; the upgrade-value criterion and this study's wording rule 2 (teleport rule as reference only). The two figures on the right are schematics.

Bilevel co-design: changes and fleet outside, orchestration inside, dual prices to rank



Chapter 5's results ladder

FLOOR	Grid evaluation of fleets and road changes
TARGET	The exchange rate and critical-link identification
CEILING	Bilevel co-design with optimality gaps

- Dual prices shrink the outer search to the binding elements, cutting the designs to simulate; the bounds and price table take about 6 min for the largest scenario and at most 25 s for the others.
- The inner level currently uses whole-route reservation (the best of the rules tested so far); a less conservative rule replaces it once ready.
- Key-element review is a fixed step: the top three dual-price elements are checked on the imagery, and the resource granularity is chosen from that before recomputing.

The vehicle-type hypothesis: large vehicles occupy even when empty; small ones can yield but run extra empty trips to team up

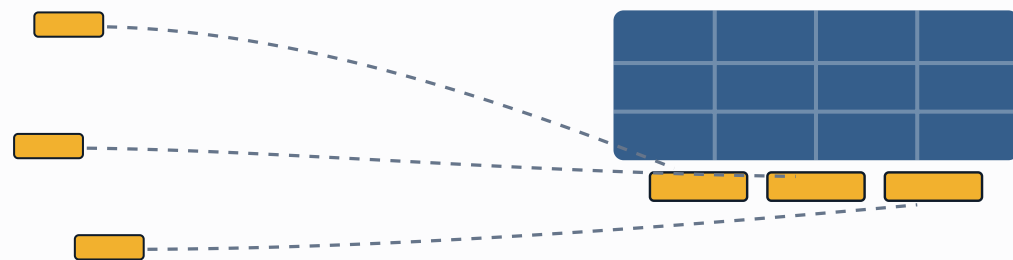
Large vehicle: holds the lane even when empty



Small vehicle: yields where passing is possible when empty



Small vehicles teamed: several per giant block, each running empty



- **Where the fleet-sizing paper starts:** with static shortest paths and no traffic, one large enough vehicle is cheaper than any coupled team for every block; prices come from 33 public tender awards. The hypothesis asks whether occupancy changes that answer.
- **Nearest same-field work (full texts):** Tao et al. 2017 compare vehicle types by simulation with road capacity counted in vehicles; Li et al. 2020 model two-vehicle cooperative transport but leave roads out. The hypothesis adds the team's footprint and road holding, the empty runs to meet and the return in convoy.
- **The case (floating docks, after the rate table):** Hanwha Ocean's 4Q24 IR lists 3 floating docks (“Floating Dock 3EA”); Okpo lifts giant blocks into them with sea cranes, modelled as a quay-side sink plus a floating-crane resource in a separate scenario. Giant blocks travel on 2–4 coupled vehicles; “about 10 giant blocks per ship” comes from another yard (Koh et al. 2011), not Okpo.

[Open the fleet-sizing paper briefing →](#)

Sources: the vehicle-type hypothesis; floating-dock model and 2–4 coupled vehicles per giant block are this study's scenario design (9 Oct 2026; four coupled: Joo et al. 2005); dock count, sea-crane lifting, giant-block count and cooperative transport checked in full text (Hanwha Ocean IR; Koh et al. 2011; Li et al. 2020); fleet-sizing result from its briefing, “single carry dominates” (static shortest paths, no traffic; 33 awards, 2022–2026). Sketches not to scale.

Established, and under way: what each task feeds

Established (checked 9–10 Oct 2026)

- **Both edges of the capacity interval (10 seeds):** the best deadlock-free rule is whole-route reservation; Okpo main [627, 1,720], erection mix [611, 703], Yantai [455, 1,252]; the dividing rule has two examples.
- **A bound free of sampling noise, with a route-free version:** system bound 1,736 at Okpo, 1,298 at Yantai; the route-free bound carries a certificate independent of solver tolerance, and a counterexample search found none.
- **Dual prices in two columns, convertible to vehicles:** one more hour a day on the dock-1 road in the erection mix ≈ 1.8 vehicles; route slack from 0 to +45.4%.
- **External anchors (full texts checked):** Okpo 339 a day; under a 24 h day field fleets sit between free flow and reservation; the closest same-field work, Tao et al. 2017, compared item by item.

1

running A less conservative deadlock-free rule (a banker's-algorithm safety check plus a step-wise feasibility search, or claims cut at the next safe holding point, per the safe-orchestration theorem); first, the Okpo map with floating docks, a fix to the direction-lock U-turn at stops and split reservation waits. → re-check of the lower edge and of the decline and safety-cost hypotheses

2

after the rule comparison Retrofit simulations; the first exchange-rate table; door granularity; capacity vs block length; the dock three-resource network. → the exchange-rate table, the prediction-vs-simulation figure, the granularity curve

3

after the rate table The floating-dock scenario; finite holding area S and $K_0(S)$. → the vehicle-type case section; the rule “what a holding area is worth in vehicles”

4

to decide Decided together after the rule comparison: new numbers for part 2 of the network-bound hypothesis, part 3's two-layer wording, and the main caliber (16 h / 24 h, daily mean / peak). → this briefing shows both scales until then

Risks and mitigations

RISK	MITIGATION
In the main scenario road changes barely move throughput	That is itself the orchestration-side reading: orchestration first; once less conservative rules and chapter 4 raise the lower edge, read the exchange rate again at the new plateau / bound
The best deadlock-free rule may change (comparison running), moving the lower edge and plateau / bound	The boundary and the rate always name the rule; the new rule joins the interval table through the same batches and the ratios are recomputed, while the dividing rule itself stays as stated
The dual is degenerate and marginal; width works in steps	Re-solved increments are in hand (the paint shop's two doors must change together; +1 m on the dock-1 road gives 0); the retrofit runs price combined changes and “widen to the next step”, relaxing one bottleneck at a time
Whether the model's vehicle counts match the field	Answered: at 24 h, Hyundai Heavy's 24 vehicles and about 30 at a large Korean yard both sit between free flow and reservation; the main caliber (16 h / 24 h, mean / peak) is fixed after the rule comparison, both reported until then
The 200 m and 25 m of segmenting are scenario settings	Imagery check: with a vehicle at the dock on the dock-1 road, the neighbouring sites stay reachable; the block-length scan (next step) turns the 25 m into a capacity–granularity curve
Gaps in public material on floating docks and giant blocks (current dock sizes, convoy speed)	Public sources give the number of floating docks and the sea-crane lifting; floating-dock flow is a separate scenario, with synthetic parts labelled as scenarios

Target journals (paper 3, chapter 5): only journals in the school's 2026 directory

Is a passing bay worth a transporter? Fleet-network co-design for road-occupying transport

FIRST CHOICE · CLASS B

- *Ocean Engineering* (class B)
- *IEEE Transactions on Automation Science and Engineering* (class B)

Shipyards transport mechanism and engineering gain (OE); network–fleet co-design and the bilevel algorithm (T-ASE)

STRETCH · CLASS A

- *Reliability Engineering & System Safety* (class A)

Condition: written as capacity loss under critical-link failure and redundancy design (the redundancy value of holding areas and second entrances)

ALTERNATIVES · CLASS B

- *Transportmetrica A: Transport Science* (class B)
- *Expert Systems with Applications* (class B)
- *Applied Soft Computing* (class B, bilevel algorithm)
- *Journal of Manufacturing Processes* (class B)

SUPPLEMENTARY · CLASS C

- *Computers & Industrial Engineering* (class C)
- *Journal of Ship Production and Design* (class C)

Class B first; class A as a stretch under the condition on its card. Graduation needs at least one paper in a class B or higher journal, and this paper aims at class B.

References (1/2)

1. Abril, M., Barber, F., Ingolotti, L., Salido, M.A., Tormos, P., Lova, A. (2008). An assessment of railway capacity. *Transportation Research Part E* 44(5), 774–806. doi:10.1016/j.tre.2007.04.001
2. Eustace, C., Bobbermen, L., Hronsky, D. (2021). Simulation of Carrapateena underground haulage operations. *Proc. 2021 Winter Simulation Conference* (industry case study).
3. Goverde, R.M.P., Corman, F., D'Ariano, A. (2013). Railway line capacity consumption of different railway signalling systems under scheduled and disturbed conditions. *J. Rail Transport Planning & Management* 3(3), 78–94. doi:10.1016/j.jrtpm.2013.12.001
4. Jensen, L.W., Schmidt, M., Nielsen, O.A. (2020). Determination of infrastructure capacity in railway networks without the need for a fixed timetable. *Transportation Research Part C* 119, 102751. doi:10.1016/j.trc.2020.102751
5. Joo, C.-M., Lee, W.-S., Lee, K.-B. [주철민, 이운식, 이강배] (2005). Transporter scheduling for block transportation in the shipyard. *Proc. 2005 KORMS/KIIE Spring Joint Conference*, 348–352.
6. Koh, S.-G., Jang, J.-H., Choi, D.-W., Woo, S.-B. [고시근, 장정희, 최대원, 우상복] (2011). Spatial scheduling for mega-block assembly yard in shipbuilding company. *IE Interfaces* 24(1), 78–86. doi:10.7232/IEIF.2011.24.1.078
7. Landex, A., Schittenhelm, B., Kaas, A.H., Schneider-Tilli, J. (2008). Capacity measurement with the UIC 406 capacity method. *WIT Transactions on The Built Environment* 103, 55–64. doi:10.2495/CR080061
8. Li, B., Jiang, Z., Tao, N., Meng, L., Zheng, H. [李柏鹤, 蒋祖华, 陶宁蓉, 孟令通, 郑虹] (2020). Ship block transportation scheduling considering cooperative transportation of flatcars. *J. Shanghai Jiao Tong University* 54(7), 718–727. doi:10.16183/j.cnki.jsjtu.2018.339
9. Na, C., Wang, Q., Han, X., Yao, J., Ma, K., Kim, Y. (manuscript in preparation). Single-vehicle capacity versus cooperative transport in the service-constrained sizing of shipyard block-transporter fleets.

References (2/2)

10. Park, C., Seo, J. [박창규, 서준용] (2006). JIBUN (location) assignment algorithm for assembly blocks: a case of Hyundai Heavy Industries. *IE Interfaces* 19(2), 160–167.
11. Reveliotis, S.A. (2000). Conflict resolution in AGV systems. *IIE Transactions* 32(7), 647–659. doi:10.1080/07408170008967423
12. Shen, H., Jeong, Y.-K., Shin, J.G., Lee, P., Woo, J.H., Lee, Y.G., Kim, S.H., Jeong, J.H. (2018). Key performance indicators and analysis method for ship block logistics flow in shipyards. *Proc. 2018 Winter Simulation Conference*.
13. Tao, N., Jiang, Z., Liu, J., Yang, Z. [陶宁蓉, 蒋祖华, 刘建峰, 杨震] (2017). Modeling and analysis of ship block transportation considering multi-type flat transportations. *Hebei J. Industrial Science and Technology* 34(6), 389–394. doi:10.7535/hbgykj.2017yx06001
14. UIC (2013). *UIC Code 406: Capacity*, 2nd ed. International Union of Railways, Paris.
15. Yim, S.-B., Roh, M.-I., Cha, J.-H., Lee, K.-Y. (2008). Optimal block transportation scheduling considering the minimization of the travel distance without overload of a transporter. *J. Society of Naval Architects of Korea* 45(6), 646–655. doi:10.3744/SNAK.2008.45.6.646

Company material: Hanwha Ocean 4Q24 and 2Q26 earnings releases (Floating Dock 3EA), checked against the originals. This study's material: its claims, hypotheses and theorems (version of 9 Oct 2026); its simulations (saturated demand, 10 seeds; the Okpo map before the floating docks and branch roads, and the Yantai map) and its system-bound, route-free-bound and dual-price computations; the draft proof of the route-free bound; the full-text checks of external anchors and literature; this study's earlier preliminary run (3 seeds); this study's research design (8 Oct 2026); the School of Shipbuilding Engineering 2026 journal directory (version of 23 Sep 2026).

Back to the whole thesis

Is a passing bay worth a transporter? First see how far the plateau sits below the bound.

<< 1

Plateau / bound far below 1: the rule is the bottleneck; orchestration first.

≈ 1

Plateau / bound near 1: the bound is tight, dual prices are the exchange rate, and fixing the road pays; afterwards the bottleneck moves to the rule (0.87 → 0.53).

Type

Does occupancy change the choice between large vehicles and teams? Tested on floating-dock transport.

← Back to the thesis briefing

Occupancy model and bounds →

Capacity interval →

Traffic orchestration →

Multi-yard test →

Fleet-sizing paper →

Thank you.